

Jaswanth Narravula

Backend engineer — identity, distributed systems and data-intensive services

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SUMMARY

I build the backend systems other applications depend on — identity providers, service boundaries, and the data layer underneath them.

At Xclusive Trading Inc. I engineered a standards-based OAuth 2.1 / OpenID Connect provider in Go that is the central single sign-on for internal applications, decomposed a production platform into isolated microservices, and cut report generation time by roughly 60% through PostgreSQL tuning.

EXPERIENCE

Software Engineer — Xclusive Trading Inc.

Present

- Engineered a standards-based OAuth 2.1 / OpenID Connect provider in Go — authorization code flow with PKCE, refresh-token rotation, RS256-signed JWTs, automated JWKS key rotation, Argon2id credential hashing and RBAC — now the central SSO provider delivering single sign-on to 3,000+ users.
- Decomposed a production platform into 6 microservices with isolated failure domains, and rebuilt its ETL workflows to be idempotent and replay-safe with exponential backoff.
- Cut report-generation time by ~60% through PostgreSQL execution-plan analysis, targeted indexing, query optimization and denormalized reporting tables over 50M+ rows, supporting 8+ production services.
- Built stateless REST APIs secured with Microsoft Entra ID SSO over OIDC, RS256 JWT validation and route-level RBAC, sustaining sub-200 ms p95 latency with per-IP rate limiting, CSRF protection, HSTS/CSP and audit logging.

Stack: Go, Python, FastAPI, Flask, PostgreSQL, Redis, Elasticsearch, React, Docker, Nginx, Microsoft Entra ID

IBM (client: DBS Bank)

- Delivered 10+ production REST APIs automating loan intake, eligibility validation and multi-stage approvals.
- Built the platform around a state-driven workflow engine with optimistic locking, so concurrent approvals could not land in invalid states.
- Reduced API response times by 30%.

Stack: Java 17, Spring Boot, JPA / Hibernate, MySQL, Redis, JUnit 5, Mockito

SELECTED PROJECTS

Enterprise SSO Identity Provider — Xclusive Trading Inc.

An OAuth 2.1 + OpenID Connect provider written from scratch in Go.

Authorization code flow with PKCE and refresh-token rotation · RS256-signed JWTs with automatic JWKS key rotation · Argon2id credential hashing and 4-tier RBAC · Admin session management and 15+ client integration examples

Full-Stack Sales Platform — Xclusive Trading Inc.

An internal sales platform in daily production use.

6 microservices with isolated failure domains · JWT role propagation from the internal SSO provider · Idempotent, replay-safe ETL with exponential backoff

Corporate Loan Processing Platform — IBM · DBS Bank

Production REST APIs for corporate loan intake and approvals.

10+ production REST APIs · State-driven workflow engine with optimistic locking · API response times reduced 30%

Workforce Management Backend — Xclusive Trading Inc.

An async-first FastAPI backend serving every internal management tier.

Azure AD single sign-on over OIDC · WebSocket notifications with heartbeat and reconnection · 4-tier route-level RBAC · Calendar scheduling with recurrence rules

Price Intelligence Platform — Xclusive Trading Inc.

An asynchronous ingestion pipeline over unreliable external sources.

Celery/Redis task orchestration with retries and failure handling · Elasticsearch fuzzy search and cross-retailer ranking · Price-history analytics

ASL Gesture Recognition — Graduate research · UAB

A PyTorch pipeline for 24 static American Sign Language gestures.

Test accuracy 13.21% -> 43.36% · Macro-F1 0.081 -> 0.351 · Transfer learning with ResNet-18

Portfolio OS — Personal project

2026

This site: five miniature operating systems that hold one career.

One pure-TypeScript kernel for windows, routing, history and persistence · A real shell engine over a virtual filesystem for the terminal · Static, accessible and fast: every URL works without JavaScript

EDUCATION

M.S. Computer Science — University of Alabama at Birmingham

Jan 2024 – Dec 2025

Database Systems · Software Design & Integration · Machine Learning · Deep Learning · Computer Vision · Network, Computer & Cloud Security

B.Tech Computer Science — Jawaharlal Nehru Technological University Hyderabad

2019 – 2023

Data Structures & Algorithms · Operating Systems · Distributed Systems · Compiler Design · Cryptography & Network Security

Credentials: AWS Cloud Architecting (Amazon Web Services), AWS Machine Learning Foundations (Amazon Web Services)

SKILLS

Languages: Go, Java 17, Python, SQL, TypeScript, JavaScript, C, C++, Bash

Backend & microservices: Spring Boot, Spring Data JPA, Hibernate, FastAPI, Flask, Node.js, SQLAlchemy, REST APIs, OpenAPI / Swagger

Identity & security: OAuth 2.1, OpenID Connect, PKCE, JWT (RS256 / JWKS), Argon2id, RBAC, Microsoft Entra ID SSO, Rate limiting, CSRF, HSTS/CSP

Data & storage: PostgreSQL, MySQL, Redis, Elasticsearch, Celery, Query plans & indexing

Frontend: React, Redux, Vite, Tailwind CSS

Cloud, tooling & CI/CD: Docker, Nginx, Linux, GitHub Actions, AWS, Azure, Maven, JUnit 5, Mockito, Postman, Selenium